

Flight Booking & Travel Data Analysis

Problem Statement:

Flight booking data contains information about airlines, routes, fares, flight characteristics, and customer booking decisions. However, raw datasets can contain missing entries, inconsistent formats, duplicate records, and values that are difficult to analyze directly.

This project focuses on turning the raw flight data into a reliable analytical dataset and using it to understand **where booking demand is concentrated, how airlines and routes perform, how fares differ, and how customers plan their travel**. Excel is used for data preparation and exploratory analysis, while Power BI is used to present the findings through an interactive dashboard.

Objective:

The project is designed to:

- Prepare a consistent and analysis-ready flight dataset.
- Detect and correct data quality issues in bookings, flights, airlines, and routes.
- Study booking volume across different airlines and travel routes.
- Compare Economy and Business class booking patterns.
- Examine differences in ticket prices across airlines, routes, and classes.
- Understand whether customers tend to book well in advance or closer to the travel date.
- Explore the relationship between flight duration, stops, and ticket pricing.
- Identify routes and airlines with notable booking activity.
- Present the results through an interactive Power BI dashboard.

Analysis:

The project investigates the following areas:

Airline Performance

- Booking volume by airline
- Economy vs Business distribution
- Average, minimum, and maximum fare by airline
- Airlines with the strongest booking demand

Route & City Patterns

- Most frequently booked routes
- Booking distribution across source and destination cities
- Average fare and duration for different routes
- Number of airlines and flights operating on routes

Fare Behaviour

- Average fare by airline and travel class
- Minimum and maximum observed fares
- Low, Medium, and High fare segments
- Variation in fares across routes

Booking Behaviour

- Last-minute vs advance booking patterns
- Booking volume across different booking windows
- Average fare associated with each booking window

Flight Characteristics

- Distribution of flight durations
- Short, Medium, and Long duration categories
- Effect of stops on flight pricing
- Comparison of direct and multi-stop travel patterns

Expected Outcome:

The completed analysis will transform the raw flight records into a **clear, interactive business intelligence dashboard**. The Power BI report will allow users to filter the data by airline, class, cities, routes, and booking window, making it easier to discover pricing patterns, popular travel routes, customer preferences, and airline performance.